Analytical Chemistry Cume

November 8, 2013

1. A time-of-flight (ToF) mass spectrometer is usually much more sensitive than a quadrupole instrument for obtaining full mass scans. What is the primary reason for this difference?

2. If you classified molecules in the following 2 ways: either as (1) organic soluble (non-polar) or (2) water soluble (polar), which classification type would be better suited for GC/MS and which better for LC/MS or is it possible to generalize at all? Explain your answer in a few sentences.

3. Which of the following mass filters is generally better suited for characterization of intact proteins: a magnetic sector instrument, a quadrupole instrument, a trapping instrument, or a time-of-flight instrument? Why?

4. Which type of instrument is known for the highest mass resolution: a magnetic sector instrument, a quadrupole instrument, a Fourier-transform instrument or a time-of-flight instrument?